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SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

I Rishikesh KS (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Rishikesh KS

89.1/100

Annexure A

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.
2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.
3. An organization has been allocated the IP block 172.16.0.0/16 for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department.

Finally, identify the remaining unused IP address range within the given network block.

4. A company uses the IP address 172.16.10.0/24 and has implemented subnetting using a /27 subnet mask for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address 172.16.10.78 belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address 172.16.10.95 falls within the same subnet range based on the given configuration.

5. A network uses the Internet Protocol version 6 address 2001:0db8:0000:0000:0000:0000:0000:0000. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of /64, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

	Problem Understanding (20%)	Method Selection (20%)	Solution Process (25%)	Accuracy of Calculation (20%)	Interpretation of Results (15%)	
Q1	2	2	2.5	2	1.5	9
Q2	2	2	2	2	1.5	9.5
Q3	2	1.5	2.5	1.5	1	8.5
Q4	1.5	2	2	1.5	1	8
Q5	2	2	2	2	1.5	9.5

11.5

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

CSA07 – COMPUTER NETWORKS

CO2 ASSESSMENT TOOL 1

ANALYTICAL PROBLEM SOLVING

Details	Info
Name	Rishiykesh
Register No.	192511466
CO2	AT1– IP addressing, subnetting and basic routing

Q1. University Network – Three Lab Subnets

A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.

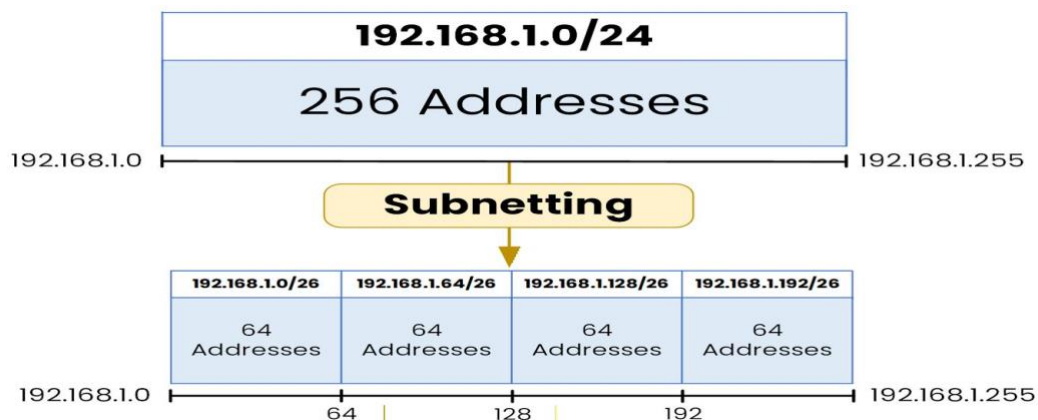
Given Data

Lab	Subnet Prefix
A	/26
B	/27
C	/28

Main network: 192.168.20.0/24

Method

- Usable IPv4 hosts = $2^H - 2$, where $H = 32 - \text{prefix length}$.



Lab A

$$H = 32 - 26 = 6 \rightarrow 2^6 - 2 = 62 \text{ hosts}$$

192.168.20.0/26

Network .0 | Hosts .1 ——— .62 | Broadcast .63

Lab B

$$H = 32 - 27 = 5 \rightarrow 2^5 - 2 = 30 \text{ hosts}$$

192.168.20.64/27

Network .64 | Hosts .65 ——— .94 | Broadcast .95

Lab C

$$H = 32 - 28 = 4 \rightarrow 2^4 - 2 = 14 \text{ hosts}$$

192.168.20.96/28

Network .96 | Hosts .97 ——— .110 | Broadcast .111

Lab	Network	Host Range	Broadcast	Usable
A	192.168.20.0	192.168.20.1–62	192.168.20.63	62
B	192.168.20.64	192.168.20.65–94	192.168.20.95	30
C	192.168.20.96	192.168.20.97–110	192.168.20.111	14

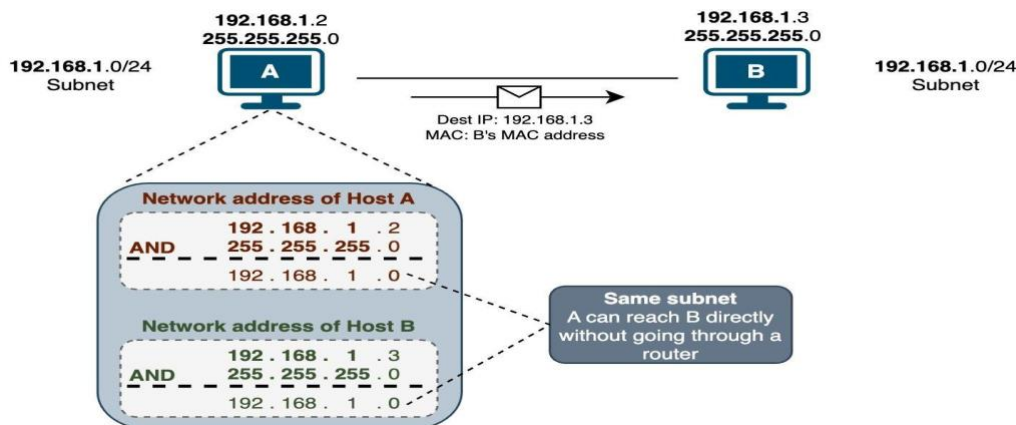
Interpretation: Separate subnets organize the labs and reduce unnecessary broadcast traffic.

Q2. Two Systems with /26

A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.

Given Data:

- System 1 = 192.168.1.130
- System 2 = 192.168.1.190
- mask = 255.255.255.192 (/26).



0-63 | 64-127 | 128-191 | 192-255

Therefore, both systems are in the same subnet and can communicate directly within the LAN without a router.

Both .130 and .190 are valid host addresses in the range .129-.190.

Step 1–Verify communication

Broadcast = 192.168.1.191.

Last host = 192.168.1.190.

First host = 192.168.1.129.

Network = 192.168.1.128.

System 2 = 192.168.1.190. Since 190 also lies between 128 and 191, it belongs to the same 192.168.1.128/26 subnet.

Step 2 – Locate System 2

Broadcast = 192.168.1.191.

Last host = 192.168.1.190.

First host = 192.168.1.129.

Network = 192.168.1.128.

System 1 = 192.168.1.130. Since 130 lies between 128 and 191, it belongs to the 192.168.1.128/26 subnet.

Step 3 – Locate System 1

The subnet ranges are 0-63, 64-127, 128-191 and 192-255.

Block size = $256 - 192 = 64$.

Step 4 – Find the /26 block size

Usable host addresses = $64 - 2 = 62$ hosts per subnet.

Total addresses = $2^6 = 64$ addresses per subnet.

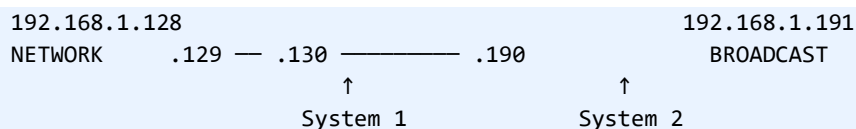
A /26 subnet leaves $32 - 26 = 6$ host bits.

255.255.255.192 = /26

Step 5 – Convert the mask to prefix length

Locate the IPs

- 130 is inside 128–191.
- 190 is inside 128–191.



Item	System 1	System 2
Network	192.168.1.128	192.168.1.128
Host range	.129-.190	.129-.190
Broadcast	.191	.191
Usable hosts	62	62

Interpretation: Both systems are valid hosts in the same 192.168.1.128/26 subnet. They can communicate directly within the LAN; a router is not required for same-subnet communication.

Q3. Department-wise Allocation

An organization has been allocated the IP block 172.16.0.0/16 for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department. Finally, identify the remaining unused IP address range within the given network block.

Given Data

Department	Prefix
HR	/25
Sales	/26
IT	/27
Admin	/28

Main network: 172.16.0.0/16

Step 1 – Understand the Allocation

The /16 network is divided sequentially according to the four given prefixes. After one subnet is assigned, the next subnet begins immediately after the previous broadcast address.

We use the rule: Usable hosts = $2^{(32 - \text{prefix})} - 2$.

Step 2 – Calculate Host Capacity

HR /25: $32 - 25 = 7$ host bits $\rightarrow 2^7 - 2 = 126$ usable hosts

Sales /26: $32 - 26 = 6$ host bits $\rightarrow 2^6 - 2 = 62$ usable hosts

IT /27: $32 - 27 = 5$ host bits $\rightarrow 2^5 - 2 = 30$ usable hosts

Admin /28: $32 - 28 = 4$ host bits $\rightarrow 2^4 - 2 = 14$ usable hosts

Step 3 – Allocate HR (/25)

Start with the first address: 172.16.0.0.

$$\text{Host bits} = 32 - 25 = 7$$

$$\text{Total addresses} = 2^7 = 128$$

$$\text{Usable hosts} = 128 - 2 = 126$$

A block of 128 addresses runs from 172.16.0.0 to 172.16.0.127.

Network = 172.16.0.0; first host = 172.16.0.1; last host = 172.16.0.126; broadcast = 172.16.0.127.

Step 4 – Allocate Sales (/26)

The next address after HR broadcast .127 is .128.

$$\text{Host bits} = 32 - 26 = 6$$

$$\text{Total addresses} = 2^6 = 64$$

$$\text{Usable hosts} = 64 - 2 = 62$$

A block of 64 addresses runs from 172.16.0.128 to 172.16.0.191.

Network = 172.16.0.128; first host = 172.16.0.129; last host = 172.16.0.190; broadcast = 172.16.0.191.

Step 5 – Allocate IT (/27)

The next address after Sales broadcast .191 is .192.

$$\text{Host bits} = 32 - 27 = 5$$

$$\text{Total addresses} = 2^5 = 32$$

$$\text{Usable hosts} = 32 - 2 = 30$$

A block of 32 addresses runs from 172.16.0.192 to 172.16.0.223.

Network = 172.16.0.192; first host = 172.16.0.193; last host = 172.16.0.222; broadcast = 172.16.0.223.

Step 6 – Allocate Admin (/28)

The next address after IT broadcast .223 is .224.

$$\text{Host bits} = 32 - 28 = 4$$

$$\text{Total addresses} = 2^4 = 16$$

$$\text{Usable hosts} = 16 - 2 = 14$$

A block of 16 addresses runs from 172.16.0.224 to 172.16.0.239.

Network = 172.16.0.224; first host = 172.16.0.225; last host = 172.16.0.238; broadcast = 172.16.0.239.

Step 7 – Find the Remaining Unused Range

Admin ends at broadcast address 172.16.0.239. Therefore, the next available address is 172.16.0.240.

The original network is /16, so it continues through 172.16.255.255.

$$\text{Unused range} = 172.16.0.240 - 172.16.255.255$$

Step 8 – Address Allocation Diagram

```
172.16.0.0/16
|
├── HR    /25 → .0 ————— .127
├── Sales  /26 → .128 ————— .191
├── IT     /27 → .192 ————— .223
├── Admin  /28 → .224 ————— .239
└── UNUSED → .240 ————— 172.16.255.255
```



Step 9 – Final Answer

Dept.	Prefix	Network	Valid Host Range	Broadcast	Usable
HR	/25	172.16.0.0	172.16.0.1–126	172.16.0.127	126
Sales	/26	172.16.0.128	172.16.0.129–190	172.16.0.191	62
IT	/27	172.16.0.192	172.16.0.193–222	172.16.0.223	30
Admin	/28	172.16.0.224	172.16.0.225–238	172.16.0.239	14

Step 10 – Decision and Interpretation

Decision: The four departments receive non-overlapping subnets with the exact prefix sizes given in the question.

Interpretation: Larger departments receive larger address blocks, while 172.16.0.240–172.16.255.255 remains available for future departments, servers, or network expansion.

Q4. Find the Subnet of 172.16.10.78

A company uses the IP address 172.16.10.0/24 and has implemented subnetting using a /27 subnet mask for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address 172.16.10.78 belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address 172.16.10.95 falls within the same subnet range based on the given configuration.

Given Data:

- 172.16.10.0/24
- new mask /27
- check .78 and .95.

Step 1 – Number of Subnets

$$\text{Borrowed bits} = 27 - 24 = 3 \rightarrow 2^3 = 8 \text{ subnets}$$

Step 2 – Usable Hosts

Host bits = $32 - 27 = 5 \rightarrow 2^5 - 2 = 30$ hosts/subnet

Step 3 – Block Size

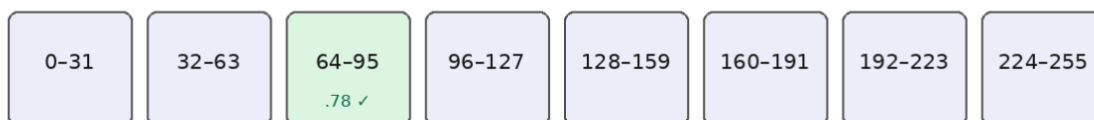
$$256 - 224 = 32$$

0-31 32-63 64-95 96-127 128-159 160-191 192-223 224-255 ↑ .78						
Item				Answer		
Subnet				172.16.10.64/27		
Network				172.16.10.64		
Valid hosts				172.16.10.65–172.16.10.94		
Broadcast				172.16.10.95		
Usable hosts				30		

Check .95: It is in the same subnet block, but it is the broadcast address and cannot be assigned to a host.

Subnet of 172.16.10.78

172.16.10.0/24 → /27



Network = 172.16.10.64/27

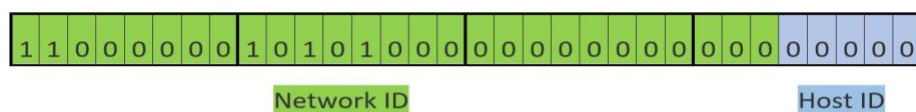
Hosts = .65 - .94 | Broadcast = .95

.95 is the broadcast address, so it is not a host address.

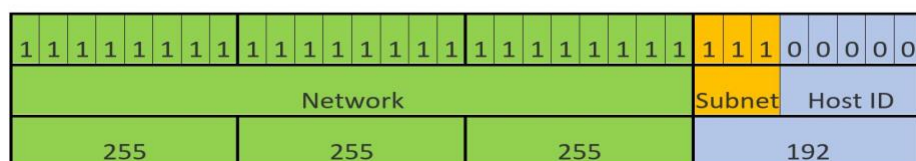
Check: 172.16.10.95 is the broadcast address of 172.16.10.64/27, so it is not a usable host address.

Interpretation: /27 divides the /24 network into 8 subnets, each with 30 usable hosts.

192.168.0.0/27 network and addresses



Subnet Mask 255.255.255.192



Q5. IPv6 Compression and /64 Prefix

A network uses the Internet Protocol version 6 address 2001:0db8:0000:0000:0000:ff00:0042:8329. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of /64, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

Given Data:

- address: 2001:0db8:0000:0000:0000:ff00:0042:8329
- prefix = /64.

Step 1 – Compress

- Remove leading zeros from each hexadecimal group.
- Replace the longest consecutive zero groups with ::.

2001:0db8:0000:0000:0000:ff00:0042:8329
→ 2001:db8::ff00:42:8329

Step 2 – Expand

2001:db8::ff00:42:8329
→ 2001:0db8:0000:0000:0000:ff00:0042:8329

Step 3 – /64 Network Prefix

64-bit Network	64-bit Host
2001:0db8:0000:0000	: 0000:ff00:0042:8329

Network prefix = 2001:db8::/64

Step 4 – Total Addresses

Host bits = 128-64 = 64 → $2^{64} = 18,446,744,073,709,551,616$

Interpretation: A /64 IPv6 network provides a very large address space for devices and future expansion.

